

GOETHE-INSTITUT PAKISTAN

Exam Slot Booking Bot

Automated • Intelligent • Instant

Prepared by **ABEER** | Delivered to **HAMZA**

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Booking a Goethe Exam Slot Manually is Nearly Impossible



Slots Vanish in Seconds

Exam slots open at unpredictable times and disappear within 2 to 5 seconds. Human reaction time simply cannot compete with an automated system.



Endless Manual Refresh

Students must sit and refresh pages for hours or days with no guarantee of success. Exhausting, unreliable, and completely unscalable.



Impossible to Scale

Running this manually for one student is painful. Doing it for multiple students at the same time is physically impossible.

A Custom-Built Automation Bot for Goethe-Institut Exam Bookings

What This Bot Does

- ✓ Monitors the Goethe-Institut portal continuously — 24 hours a day, 7 days a week
- ✓ Detects available exam slots the millisecond they appear
- ✓ Books slots automatically with zero human input required
- ✓ Handles multiple students simultaneously in parallel
- ✓ Runs silently on a cloud server — your PC can be completely off
- ✓ Logs every single action with full timestamps for a complete audit trail



The Result

Slot Booked
Instantly

Before Any Human Can React

Technology Stack — Built with Production-Grade Tools



Python 3

Core language powering all bot logic. The industry standard for automation and web scraping — fast, reliable, and battle-tested.



Selenium

Browser automation engine. Controls a real Chrome instance and interacts with the Goethe portal exactly the way a human user would.



Flask

Lightweight web framework that runs the control dashboard. Manage the bot and monitor live status from any browser.



Docker

Containerizes the entire system. One single command deploys the bot anywhere — no dependency headaches, no complex server setup.

How All Components Connect



 All components run on VPS (Oracle Cloud Free / Hetzner) • Domain managed via Cloudflare • HTTPS auto-secured

 Hamza accesses the dashboard from any browser, any device, via custom domain

Part 1 of 2 — Startup, Login & Monitoring

01

Launch the Dashboard

Hamza opens the Flask web interface in any browser. Enter student credentials, exam type, and target date via the control panel.

02

Configuration Loaded

Bot reads the .env file for login credentials and exam preferences. Validates all inputs before doing anything else.

03

Headless Browser Launches

Selenium fires up an invisible Chrome browser on the VPS. No screen needed — it runs silently in the server background.

04

Navigate to Goethe Portal

Bot opens goethe.de, navigates to the Pakistan exam booking section, and automatically handles login and cookie consent prompts.

05

Monitoring Loop Begins

Bot enters an aggressive polling loop — continuously refreshing the exam slot page at randomized intervals of 0 to 800ms, 24 hours a day.

Part 2 of 2 — Detection, Booking & Confirmation

06

Slot Detected

Bot scans for the SELECT MODULES button or any enabled booking button. Goethe marks unavailable slots as disabled — bot reacts the moment that attribute is removed.

07

Instant Click Fired

Bot clicks the booking button in milliseconds — far faster than any human. If a click fails or times out, bot auto-restarts the entire session from Step 1.

08

Forms Auto-Filled

Bot automatically fills in all required fields — student name, ID number, contact details, exam module preferences. Zero manual typing required from Hamza.

09

Booking Confirmed

Bot completes the full multi-step booking flow, handles any confirmation or verification screens, and secures the exam slot successfully.

10

Logged and Reported

Every page load, click, form fill, and booking outcome is written to the log file with timestamps. Full audit trail for every student session.

Anti-Detection and Intelligent Polling System

How the Bot Stays Safe

R

Randomized Timing

Refreshes at random 0 to 800ms intervals. No fixed pattern means no bot-detection signature can be flagged on Goethe servers.

V

Reload Verification

Only refreshes after confirming the previous page loaded successfully. Prevents request flooding and IP throttling.

A

Auto-Restart on Failure

Any step fails? Bot automatically restarts the full session from the beginning. Zero downtime, zero manual intervention needed.

P

Privacy Prompt Handling

Re-accepts cookies and privacy consent on every restart — exactly mimicking normal human browser behavior.

Polling Loop Logic

1

Check for SELECT MODULES text

2

Check for enabled booking button

3

Neither found? Wait 0 to 800ms

4

Reload page — confirm RELOAD_OK

5

Re-check privacy/cookie prompts

✓

Slot found — proceed to booking!

Multi-Student Mode — Built for Your Agency

How Multiple Students Are Managed

C

All student data stored in a structured CSV file — name, email, password, exam type, and preferred date

S

Bot reads each student row and launches a dedicated browser session per student, one by one or in parallel

P

Each session runs independently — multiple bookings can be attempted simultaneously on the same VPS server

L

Separate log entries per student — full tracking of who got which slot and exactly when it was booked



students.csv

name	email	exam	date
Student 1	s1@mail.com	B1	Aug 1
Student 2	s2@mail.com	A2	Aug 1
Student 3	s3@mail.com	B2	Aug 5
Student 4	s4@mail.com	A1	Aug 5
Student 5	s5@mail.com	C1	Aug 8

Each row = one complete student booking session

Deployment and Hosting — Ready to Go Live



VPS Server

Oracle Cloud Free / Hetzner

Bot runs 24/7 on a cloud server. Never needs your PC running. Handles everything autonomously.



Docker Deploy

One-Command Setup

`docker-compose up -d` is all it takes. Bot, dashboard, and all dependencies spin up automatically.



Cloudflare Domain

Custom Domain + HTTPS

Custom domain points to VPS via Cloudflare. SSL certificate managed automatically. Access from anywhere.



.env Security

Secure Credential Storage

All passwords and keys stored in `.env` file — never hardcoded in source code. Swap credentials anytime safely.

Hamza's Complete Delivery Package



Full Source Code

Complete Python + Flask codebase.
Private GitHub repo, structured and
documented.



Docker Setup

docker-compose.yml and Dockerfile
pre-configured. One command to fully
deploy.



.env.example

Template for all environment variables.
Fill in credentials — done.



requirements.txt

All Python dependencies pinned. pip
install -r gets everything running
instantly.



README Guide

Full setup and deployment
documentation. Deploy in under 30
minutes.



License File

Proper LICENSE included. You own the
right to run this bot for your agency.



You're Ready, Hamza.

The bot is built. The code is delivered. Your agency is ready to launch.

10

Automation Steps

< 3s

Time to Book a Slot

24/7

Always Monitoring

∞

Scalable Students